

2024 Bi H1 Q2

Section: DNA and the Genome

Topic: Organisation of DNA

Question Summary:

How is DNA organised in prokaryotes?

Worked Solution:

- Prokaryotes do not have a nucleus; their genetic material is in the cytoplasm.
- They have a single circular chromosome (not linear) plus extra small circular DNA called plasmids.
- Therefore the correct description is circular chromosomes and plasmids in the cytoplasm.

Final Answer: B (circular chromosomes and plasmids in the cytoplasm).

Revision Tips:

- Prokaryote: circular chromosome + plasmids; no nucleus.
- Eukaryote: linear chromosomes in a nucleus; plasmids are rare (but present in some yeasts).
- Plasmids often carry genes for advantages such as antibiotic resistance.